



OPEN AHP multi criteria analysis for landslide susceptibility mapping in the Tellian Atlas chain

Karim Zighmi^{1,2}✉, Farid Zahri^{1,2}, Khadeijah Faqeih³, Afaf Al Amri⁴, Hadji Riheb^{1,2}, Somayah Moshfir Alamri³ & Eman Alamery³

Algeria's Tellian Atlas, characterized by steep topography and complex geology, is particularly susceptible to landslides, necessitating robust hazard assessment frameworks. This study develops a landslide susceptibility map (LSM) using a GIS-based multi-criteria approach integrating statistical evaluation and expert judgment. A detailed landslide inventory, comprising 501 documented events, was established through high-resolution satellite imagery, aerial photographs, and field verification. Eleven conditioning factors were considered (slope angle, aspect, elevation, curvature, lithology, precipitation, and distances to faults, rivers, and roads, as well as stream power index (SPI) and topographic wetness index (TWI)) derived from remote sensing data, geological maps, and meteorological records. These variables were standardized and analyzed using the analytical hierarchy process (AHP), which generated relative weights through pairwise comparisons. A multicollinearity analysis was conducted to ensure statistical robustness, with factors displaying variance inflation values below critical thresholds. The resulting LSM was validated using a receiver operating characteristic (ROC) curve, yielding an area under the curve (AUC) value of 0.75, indicating good predictive performance. The findings reveal that areas with steep slopes, clay-rich lithologies, and proximity to tectonic features exhibit the highest susceptibility. The LSM offers a valuable tool for spatial planning, early warning systems, and risk mitigation strategies, particularly in the face of increasing climatic extremes that intensify landslide triggers. This integrative approach not only enhances geohazard management in mountainous terrains but also provides a replicable framework for similar environments globally, contributing to the broader goal of sustainable land use and disaster resilience.

Keywords Analytical hierarchy process, Geospatial database, Multicollinearity analysis, Slope stability factors, Topographic wetness index

Landslides are frequent geological hazards that cause significant damage to communities, particularly in subtropical regions. As socioeconomic development accelerates and human activities increasingly encroach upon natural landscapes, the incidence of landslides has risen, posing serious threats to lives and hindering economic progress in hilly and mountainous areas. Producing accurate and dependable landslide risk maps is therefore essential for local authorities and planners to make informed decisions about land use and to implement effective disaster mitigation strategies that minimize potential losses^{1–6}.

In recent years, northeastern Algeria has witnessed a marked increase in landslide activity, primarily driven by a combination of natural and human-induced factors. Intense rainfall events, seismic disturbances, and unsustainable land use practices (such as inadequate urban planning and poor construction methods) have all contributed to this rising hazard⁷. Several high-profile incidents underscore the gravity of the issue. For instance, a 2018 landslide in the city of Constantine led to the collapse of a building and left several individuals injured⁸. In 2019, the province of Boumerdes injured considerable infrastructure damage following a major landslide event⁹. More recently, the 2021 earthquake in Mila Province triggered multiple landslides that severely disrupted local

¹Department of Earth Sciences, Institute of Architecture and Earth Sciences, Ferhat Abbas University, Setif 19137, Algeria. ²Laboratory of Applied Research in Engineering Geology, Geotechnics, Water Sciences, and Environment, Ferhat Abbas University, Setif, Algeria. ³Department of Geography and Environmental Sustainability, College of Humanities and Social Sciences, Princess Nourah bint Abdulrahman University, P.O. Box 84428, 11671 Riyadh, Saudi Arabia. ⁴Department of Geography, College of Humanities and Social Sciences, King Saud University, 11451 Riyadh, Saudi Arabia. ✉email: zighmi.karim19@gmail.com

infrastructure and economic activity¹⁰. A particularly tragic rockfall occurred in Aokas in 2015, resulting in loss of life and widespread property destruction^{11,12}.

Beyond their immediate impact on lives and property, landslides also impose long-lasting economic and environmental burdens. These include soil degradation, deforestation, and disruption of local ecosystems, which collectively undermine regional ecological balance and agricultural productivity¹³. As a result, there is an urgent need to develop comprehensive landslide susceptibility maps and implement effective risk mitigation strategies, especially in the Mediterranean context, where Algeria is particularly vulnerable.

To address this challenge, the scientific community has increasingly turned to advanced methodologies for landslide hazard analysis. The advent of geospatial technologies (such as Remote Sensing (RS), Geographic Information Systems (GIS), Global Positioning Systems (GPS), and high-performance computing) has significantly enhanced the ability to gather, process, and interpret geo-environmental data. These innovations have enabled a paradigm shift from traditional, expert-driven qualitative assessments to modern, data-intensive quantitative models, thereby improving the accuracy and consistency of landslide predictions.

A variety of modeling approaches have been adapted to this end¹⁴. Statistical techniques, including logistic regression, artificial neural networks, fuzzy logic, and decision trees, are commonly applied to explore and quantify the relationships between landslide occurrences and influencing variables^{15–18}. Concurrently, machine learning methods (such as Random Forest (RF) and Support Vector Machines (SVM)) have gained traction for their capacity to uncover complex patterns within large datasets, thereby enhancing predictive performance. In parallel, expert systems continue to play a vital role by incorporating domain-specific knowledge to assess susceptibility through integrated qualitative and quantitative criteria.

Numerous studies have focused on landslide susceptibility, hazard assessment, and risk evaluation, particularly in mountainous regions such as the Himalayas. Notably, Saha et al.^{19,20} have made significant contributions through the development and evaluation of various GIS-based and machine learning models for landslide susceptibility mapping. Their work includes a novel hybrid machine learning approach applied to the Darjeeling Himalayas, as well as a multi-criteria decision analysis (MCDA) framework tailored for a section of the Darjeeling District in the northeastern Himalayas of India. They have also investigated the effectiveness of integrated models combining Artificial Neural Networks (ANN), Fuzzy Analytic Hierarchy Process (Fuzzy-AHP), and MCDA techniques for mapping landslide-prone areas in the Darjeeling region of West Bengal. In addition, their research has extended to the Siwalik Himalayan Zone, where they explored the utility of both statistical and machine learning methods for landslide susceptibility assessments using geospatial tools. Complementing these efforts, Dey et al.²¹ have examined the role of uncertainty in GIS-based landslide susceptibility models, emphasizing the implications of incorporating machine learning algorithms within the complex terrain of the Darjeeling Himalayas.

This study uses semiquantitative approach to evaluate landslide susceptibility in the middle Seybouse watershed. The AHP method, developed by Saaty²², facilitates pairwise comparisons of decision variables, yielding a relative dominance value that aids in the qualitative and quantitative prioritization of factors²³.

While AHP is adept at managing complex criteria and offers adaptability across analytical scales, it demands substantial expertise and comprehensive data, which can be resource-intensive. The primary objective of this research is to assess landslide susceptibility in Northeast Algeria by integrating AHP model, considering geological, topographical, hydrological, and anthropogenic factors. Furthermore, the effectiveness of the model will be evaluated to assess its performance. The study encompasses three key stages: standardization of variables, factor weighting, and criteria grouping. The findings will contribute valuable insights into the susceptibility of the Northeast region to landslides, facilitating the development of effective risk management strategies to mitigate future impacts.

The primary novelty of our work lies in the integration of a comprehensive, multi-source landslide inventory with a robust GIS-based AHP modeling framework, tailored specifically to the complex geomorphological and lithological setting of the Tellian Atlas. Unlike many previous studies that rely on limited or non-validated inventories, our approach employs an extensive field-verified database of 501 landslides and incorporates a stratified sampling design to reduce spatial bias. The methodological rigor (evident in the multicollinearity diagnostics and consistent data harmonization) ensures that each conditioning factor independently contributes to the final susceptibility model, enhancing its reliability and regional applicability.

Study area

The Middle Seybouse watershed, located in northeastern Algeria, encompasses an area of over 1400 square kilometers and is part of the mountainous landscape of the Tellian Atlas (Fig. 1). This region exhibits a complex structural framework shaped by significant tectonostructural activity resulting from the collision between the African and Eurasian tectonic plates, which has led to the formation of thrusts and folds^{24,25}.

Topographically, the watershed features hills and mountains with elevations ranging from 100 m in Seybouse valley to 1404 m, at Jebel Mahouna peak. Erosion has notably influenced the landscape, creating moderate slopes and rounded summits. The climate is semi-arid, with annual rainfall reaching up to 800 mm²⁶. The Seybouse main river serves as the primary hydrological feature in the region.

Geologically, the area consists of carbonate formations from the Jurassic to Quaternary periods. Triassic diapiric extrusions of plastic clay and gypsum are found in the northern Hammam N'bail region. The eastern side of Jebel Debagh features alternating layers of marl and shale, while Jurassic formations are marked by black dolomites and thin calcareous beds. The Upper Cretaceous period is characterized by marls and dolomitic limestone with abundant pelagic microfauna, while the Lower Cretaceous includes Barremian black marls rich in pyritic ammonites. The Albian period features black to greenish marls and micritic limestone. The Numidian thrust sheet consists mainly of sandstone and clayey intercalations, whereas the Tellian thrust sheet comprises allochthonous layers from the Cretaceous-Paleogene era. The Penthivère flysch, spanning the Cenomanian

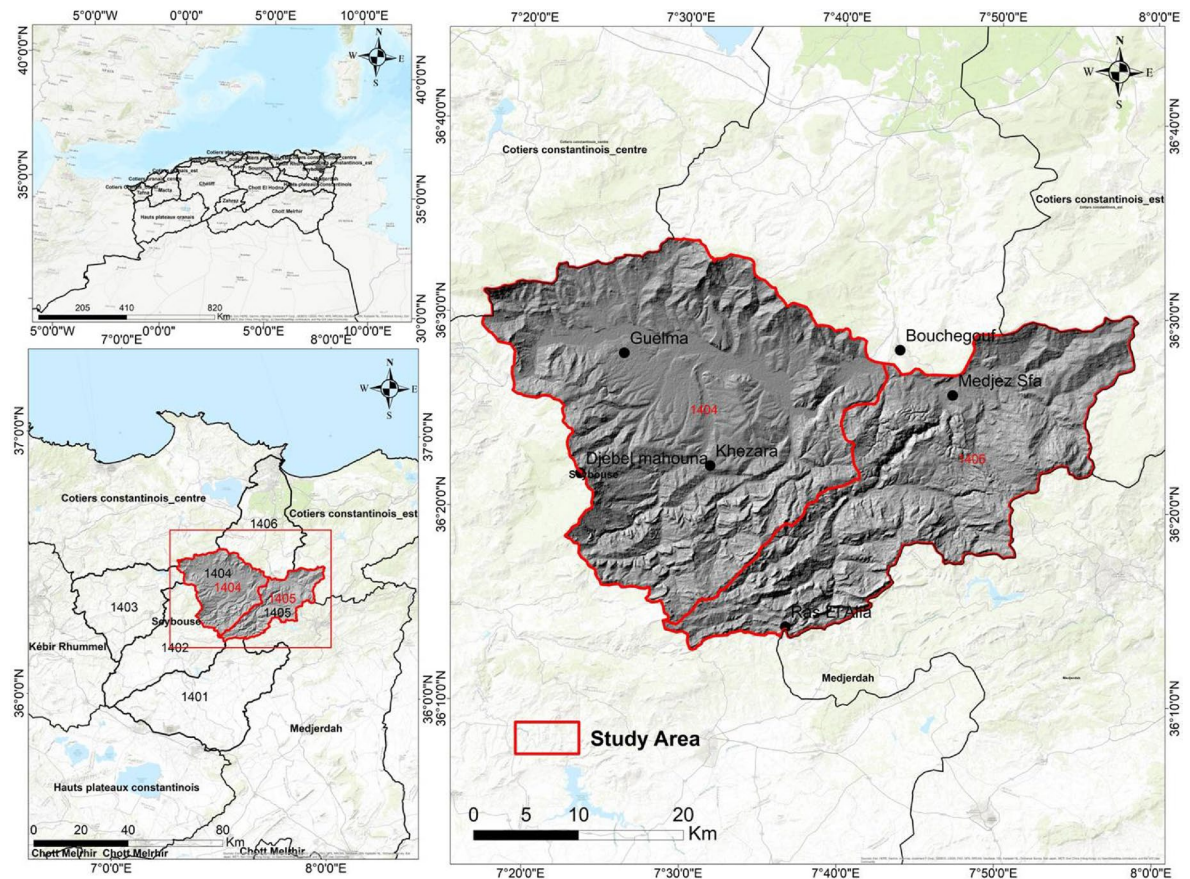


Fig. 1. Geographical location of the study area.

to Oligocene periods, is located southeast of Jebel Debagh, while the Guerouche flysch to the south consists primarily of sandstone with clay-schist intercalations. The Mio-Pliocene period is represented by sandy clays and red conglomerates, and the Quaternary period includes alluvial terraces of gravel, silt, clays, and conglomerates.

Materials and methods

Data sources

A geospatial database was developed, integrating various thematic data layers that represent key landslide conditioning factors. These factors include slope gradient, slope aspect, elevation, lithology, distance from faults, rivers, and roads, as well as SPI and TWI. Additionally, triggering factors such as rainfall were incorporated into the database, as they significantly influence landslide occurrence. To create this spatial database, data were sourced from multiple platforms. DEM data provided essential information on slope angle, aspect, curvature, elevation, and stream networks. Furthermore, eight geological maps of 1:50,000 scale were digitized to integrate lithological and tectonic features. Satellite imagery coverage, was employed to accurately identify landslide locations. The methodology comprised three key stages: variable standardization, factor weighting, and criteria grouping. For statistical analysis, Excel Stat Pro software was utilized, while ArcGIS 10.8 was employed for the geoprocessing.

Landslide inventory map

The assessment of landslide susceptibility begins with developing a comprehensive landslide inventory map, which forms the basis for identifying areas at risk of future events. This map documents the spatial distribution of past landslides, offering insights into their relationships with environmental and human-induced factors. The critical role of such inventories in landslide analysis is well established in the literature^{27–30}.

Accurate delineation of the location and extent of historical landslides is vital for producing reliable susceptibility maps. This involves recording details such as timing, type, volume, and movement characteristics of landslides within the study area. Our approach combined visual interpretation of high-resolution satellite imagery, aerial photographs, and extensive field verification, supplemented by a thorough review of historical records and scientific publications³¹.

In this study, 501 landslides were identified through a multi-source methodology integrating remote sensing data, field observations, and existing literature. These landslides are mainly concentrated along steep slopes, road cuts, riverbanks, and escarpments, with common triggers including erosion-induced slope instability, steep gradients, and sparse vegetation, especially on cliff faces.

Given the complex geomorphology and lithological diversity of the Tellian Atlas, a multi-source approach was essential to ensure the inventory's completeness and spatial representativeness. This involved interpreting high-resolution satellite images, analyzing aerial photographs, consulting historical archives, and conducting extensive field campaigns supported by GPS-based ground truthing (Fig. 2). Fieldwork focused on systematically documenting landslides across all major lithostratigraphic units, with special attention to highly susceptible formations such as Triassic clayey-gypso-sandstone facies, Quaternary sediments, Numidian clays, black marls, marly limestones, red clays, puddingstone sandstones, lacustrine limestones, and schistous marls.

To reduce spatial sampling bias (often caused by uneven terrain accessibility or mapping resolution) a stratified sampling strategy was employed. The spatial distribution of landslides was further validated using GIS overlays of lithology and terrain classifications. This integrative approach ensured a balanced, representative dataset, thereby improving the reliability, generalizability, and predictive accuracy of the subsequent susceptibility modeling.

Landslide-conditioning factors

The selection of the eleven landslide-conditioning factors was based on a combination of scientific relevance, data availability, and their demonstrated influence on slope instability within the study area. The factors were chosen following an extensive literature review and validated through local geomorphological and hydrological analyses. Each factor is commonly recognized in landslide susceptibility studies and contributes uniquely to the spatial variability of landslide occurrence.

In this landslide susceptibility study, a diverse datasets are employed to effectively represent the key factors influencing slope stability across the landscape. The slope angle, a primary control on slope failure, was derived from a digital elevation model (DEM) and systematically divided into five equal intervals to support detailed spatial analysis. Complementing this, slope aspect (the compass direction that slopes face) was automatically classified into nine distinct categories, reflecting its well-established impact on landslide distribution. Plane curvature, which governs surface water flow and erosion processes, was extracted from the DEM using GIS techniques and categorized into concave, convex, and flat forms to capture subtle geomorphological variations. Elevation data, spanning from 95 to 1,404 m above sea level, were segmented into five uniform intervals to incorporate altitudinal influences on landslide susceptibility. Geological faults, mapped at a 1:50,000 scale, were integrated by calculating distances within 1,000-meter buffer zones, acknowledging their role in weakening rock masses and facilitating slope failures. Hydrological factors were addressed through proximity analysis to rivers,

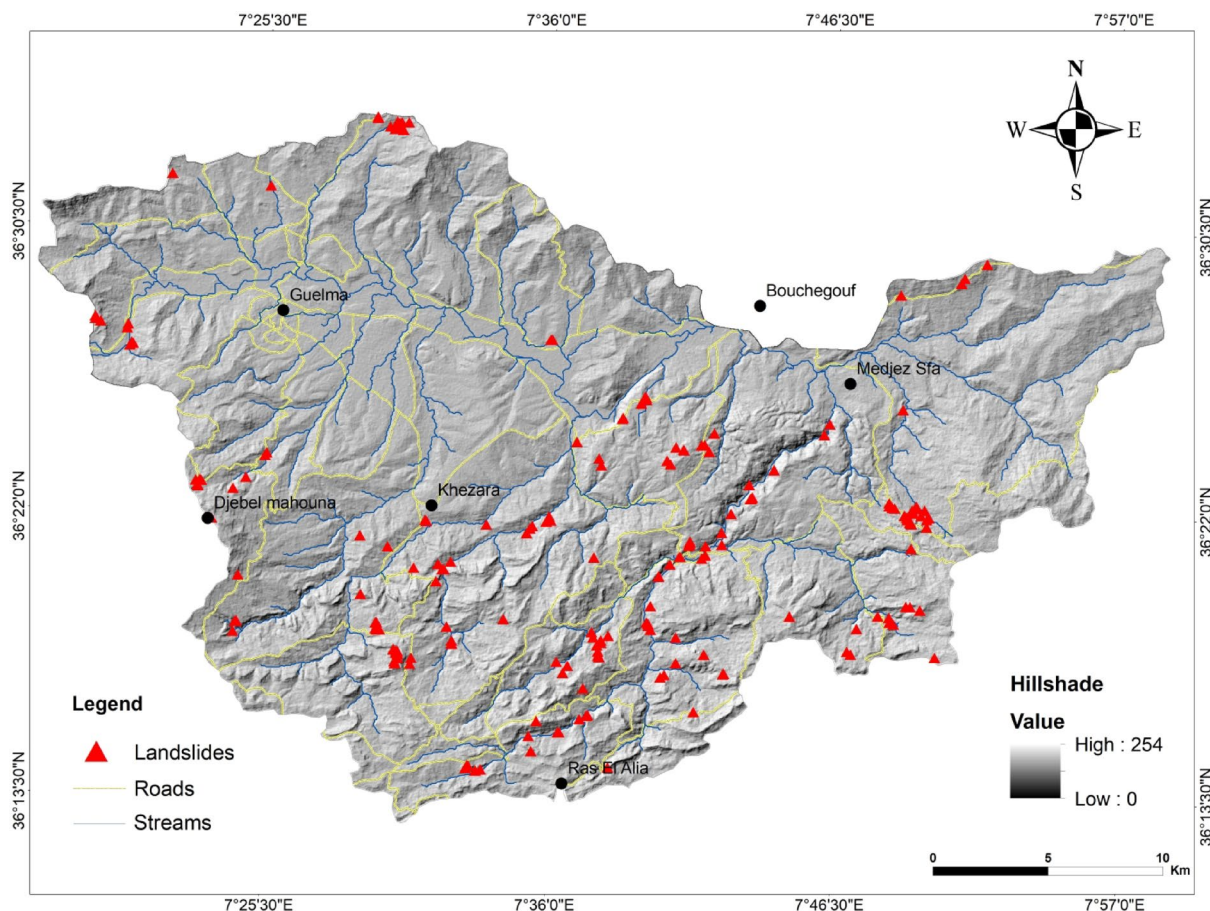


Fig. 2. Landslide inventory map of the study area.

delineated by five buffer zones, and by calculating the Stream Power Index (SPI), which combines catchment area and slope gradient to quantify the erosive force of flowing water. Soil moisture conditions, critical for slope stability, were estimated using the Topographic Wetness Index (TWI), derived from upslope contributing area and local slope angle. Human-induced effects were considered by mapping distances to roads with 500-meter buffer zones, recognizing the destabilizing impact of infrastructure development. Lithological diversity was captured through detailed geological maps, classifying rock and soil units from soft clayey deposits to more resistant hard rock formations. Lastly, precipitation patterns, a key trigger of landslides, were represented by an isohyets map generated through interpolation of rainfall measurements, illustrating spatial variability in rainfall intensity. Together, these datasets integrate geomorphological, geological, hydrological, climatic, and anthropogenic dimensions, forming a comprehensive and scientifically sound basis for modeling landslide susceptibility within the study area (Table 1).

Slope angle is a critical factor in assessing slope stability and is commonly incorporated into landslide susceptibility models³². In this study, a slope angle map was derived from a DEM and classified into five equal intervals (Fig. 3).

Slope aspect, which indicates the direction a slope faces, significantly influences landslide occurrence³³. In our analysis, the slope aspect was systematically categorized into nine directional classes (Fig. 4).

Plane curvature affects water flow dynamics, influencing erosion patterns³⁴. Using ArcGIS 10.8 processing, plane curvature was derived from the DEM and classified into three categories: concave, convex, and flat (Fig. 5).

Elevation is an important factor in landslide susceptibility assessments. The study area has altitudes ranging from 95 to 1,404 m above sea level, which were classified into five equal intervals of 230 m to facilitate analysis (Fig. 6).

Faults can trigger landslides by causing tectonic ruptures that weaken rock formations³⁵. Faults were identified from 1:50,000 geological maps, and the distance to these faults was calculated using a buffer approach with 1,000-meter intervals (Fig. 7).

In mountainous areas, rivers play a crucial role in landslide initiation through runoff³⁶. To assess their influence, five buffer zones were established around watercourses (Fig. 8).

Human activities, particularly road construction, significantly contribute to landslide occurrences³⁷. The impact of roads on slope stability was examined by creating five buffer zones at 500-meter intervals based on proximity to roads (Fig. 9).

The SPI quantifies the erosive potential of a river or stream, which is closely linked to landslide occurrence in mountainous regions³⁸. The SPI was classified into five categories (Fig. 10), calculated using the formula:

$$SPI = As \times \tan \beta \tag{1}$$

With “A”s is the specific catchment area and “b” is the local slope gradient (°).

The TWI measures soil moisture retention based on topographical features, influencing landslide susceptibility³⁹. TWI values were classified into five categories (Fig. 11) using the Eq. (2):

$$TWI = \ln (\alpha / \tan \beta) \tag{2}$$

With “α” is the cumulative upslope area through a point and tan “β” is its angle of slope.

Lithology is a key consideration in landslide stability analyses. Geological maps at a scale of 1:50,000 were used to create a lithological map, highlighting various units, including soft clayey facies and hard rock facies (Fig. 12).

Precipitation is a major trigger for landslides. Excess water increases pore water pressure in soils; reducing their stability. To assess precipitation impacts, an isohyets map was generated by interpolating point precipitation data

Landslide-conditioning factor	Data source/description	Classification/processing	Notes/relevance
Slope angle	Derived from DEM	Classified into 5 equal intervals (°)	Critical for slope stability assessment
Slope aspect	Derived from DEM	Categorized into 9 directional classes	Influences landslide occurrence by slope exposure
Elevation	DEM-based altitude data	Classified into 5 equal intervals (e.g., 230 m)	Important for landslide susceptibility; affects climate and vegetation
Plane curvature	Derived from DEM using GIS tools (ArcGIS)	Classified into concave, convex, flat	Affects water flow and erosion patterns
Lithology	Digitalized from Geological maps (1:50,000 scale)	Classified into various rock/soil units	Key for rock strength and stability analysis
Precipitation	Isohyets map from interpolated point rainfall data	Contour lines representing equal precipitation	Major landslide trigger by increasing soil pore pressure
Distance to faults	Geological maps with fault lines	Buffer zones at 1000 m intervals	Faults weaken rock formations
Distance to rivers	Hydrographic maps	Buffer zones (5 zones) around rivers	Rivers influence landslides via runoff and erosion
Distance to roads	Road network maps	Buffer zones at 500 m intervals	Roads contribute to slope instability due to construction
Stream power index (SPI)	Calculated from DEM: $SPI = As \times \tan \beta$	Classified into 5 categories	Quantifies erosive potential of streams related to landslides
Topographic wetness index (TWI)	Calculated from DEM: $TWI = \ln(\alpha / \tan \beta)$	Classified into 5 categories	Measures soil moisture retention influencing landslide risk

Table 1. Landslide-conditioning factors: data sources, and classification methods.

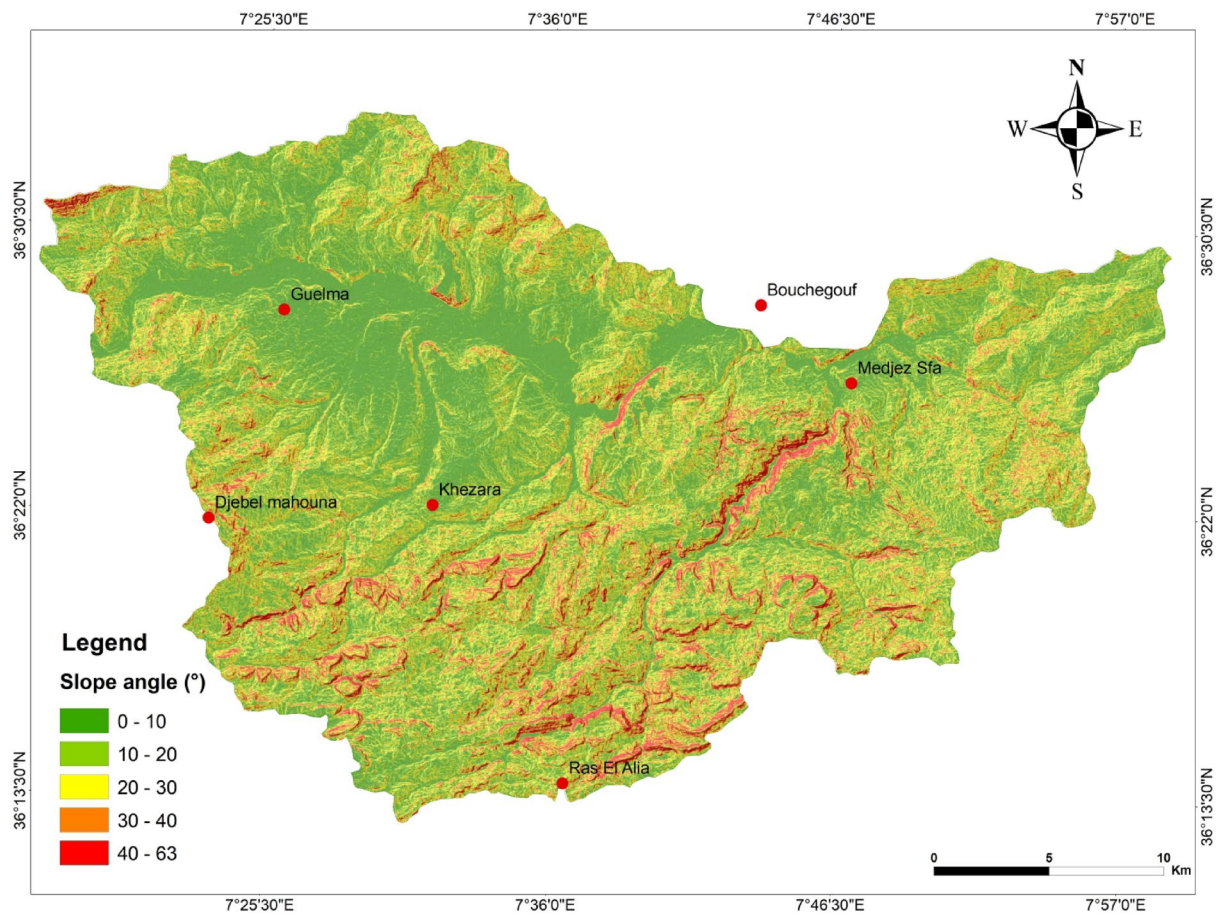


Fig. 3. Slope map (in degree) of the study area.

to estimate levels across the study area. Contour lines were then drawn to connect points of equal precipitation (Fig. 13).

To achieve spatial consistency and data harmonization during GIS-based processing, all raster layers were resampled to a uniform resolution of 30×30 m. This resolution was chosen to match the Shuttle Radar Topography Mission (SRTM) Digital Elevation Model (DEM), which served as the reference layer throughout the analysis.

Analytical hierarchy process model

The Analytical Hierarchy Process (AHP), developed by Saaty in the 1980s, is a robust multi-criteria decision-making (MCDA) method widely used in environmental modeling, spatial planning, and risk assessment. AHP is particularly suited for landslide susceptibility mapping as it allows the integration of both quantitative data and qualitative expert judgment to evaluate complex, multi-factorial problems.

At the core of AHP lies the construction of a pairwise comparison matrix, where the relative importance of contributing factors is assessed using Saaty's fundamental scale ranging from 1 to 9. A score of 1 indicates equal importance between two criteria, while higher values express a stronger preference for one factor over another. The matrix is square and reciprocal, with diagonal elements assigned a value of 1, indicating self-comparison. The outcome of this systematic comparison yields a weight for each factor, reflecting its influence in the hierarchical decision framework.

The final landslide susceptibility map (LSM) is calculated using a weighted linear combination of the selected factors, as represented by the Eq. (3):

$$\text{LSM} = w_1F_1 + w_2F_2 + \dots + w_nF_n \quad (3)$$

where w_1 to w_n are the normalized weights assigned to each landslide conditioning factor F_1 to F_n , respectively. These weights are derived from expert judgment, empirical analysis, and a review of relevant literature, supported by field-based observations tailored to the study area's geotechnical context.

To ensure methodological rigor and minimize subjectivity, the Consistency Index (CI) and the Consistency Ratio (CR) are calculated. These metrics assess the logical coherence of the pairwise comparisons. The CI is computed as:

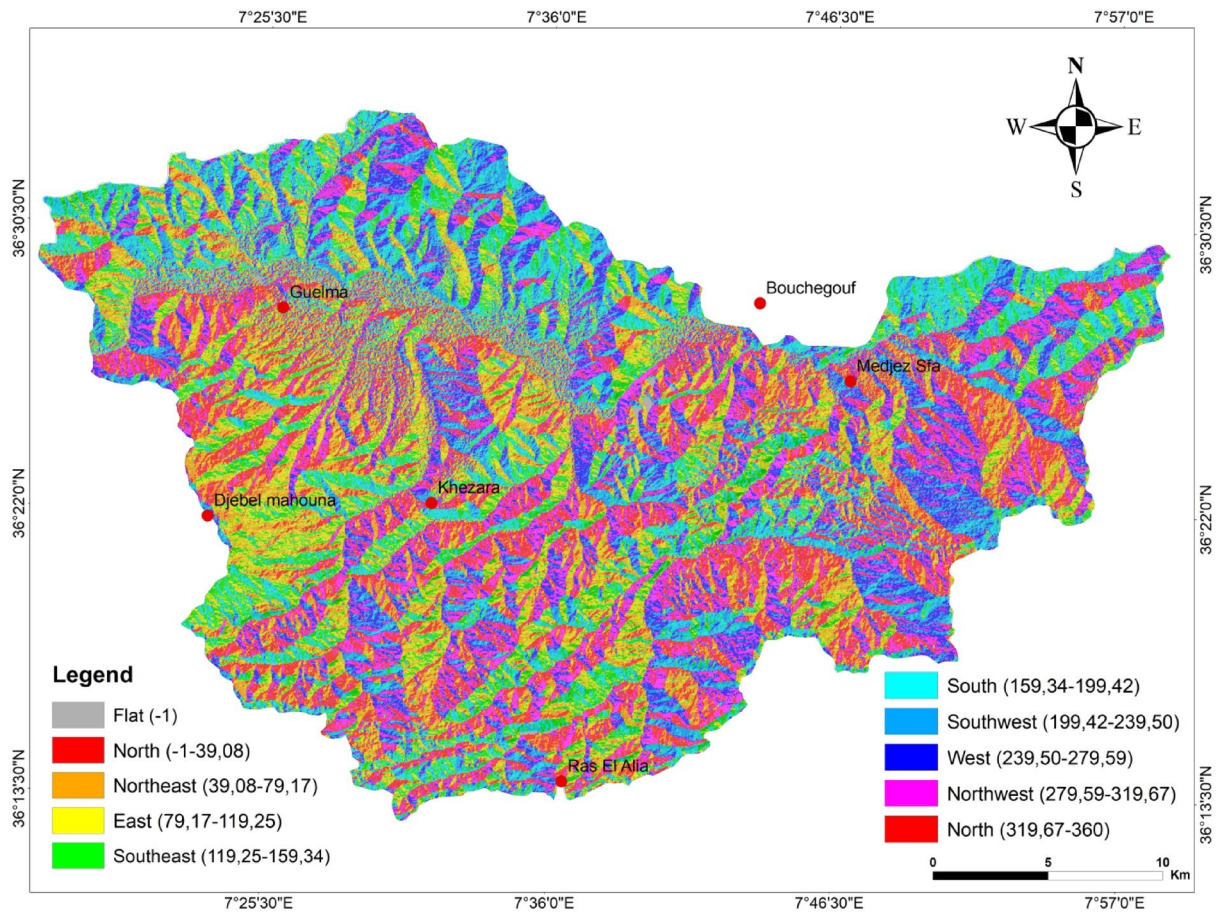


Fig. 4. Slope aspect map of the study area.

$$CI = \frac{\lambda_{max} - n}{n - 1} \quad (4)$$

where λ_{max} is the principal eigenvalue of the matrix and n is the number of factors. The CR is then determined by:

$$CR = CI/RI \quad (5)$$

with RI representing the Random Index, a benchmark value dependent on the matrix size. A CR value less than 0.10 is generally considered acceptable, indicating that the judgments made in the matrix are consistent and that the resulting weights are reliable.

AHP's strength lies in its structured yet flexible framework that accommodates both subjective and objective assessments. This makes it an effective decision-support tool for landslide susceptibility mapping, capable of integrating empirical measurements with expert insight to produce spatially accurate and scientifically credible risk assessments.

Before integrating the variables into the Analytic Hierarchy Process (AHP) model, a multicollinearity analysis was conducted to ensure the statistical independence of the conditioning factors. This step is critical to prevent the inclusion of highly correlated variables, which could distort the weighting process and compromise the interpretability of the model results.

Multicollinearity was assessed using the Variance Inflation Factor (VIF) and Tolerance (calculated as $1/VIF$), applied to standardized raster datasets representing the conditioning factors. As a general rule, VIF values exceeding 10 signal problematic multicollinearity, while Tolerance values below 0.1 suggest a high degree of redundancy.

This analysis, (Table 2), revealed VIF values ranging from 1.23 to 3.85 and corresponding Tolerance values exceeding 0.25. These results confirm that multicollinearity among the selected variables remains within acceptable thresholds, indicating that each factor provides distinct and meaningful input to the model.

By minimizing redundancy and ensuring variable independence, this analysis reinforces the methodological integrity of the AHP-based landslide susceptibility mapping. Consequently, the derived factor weights are more likely to reflect true geomorphological influences rather than statistical distortions.

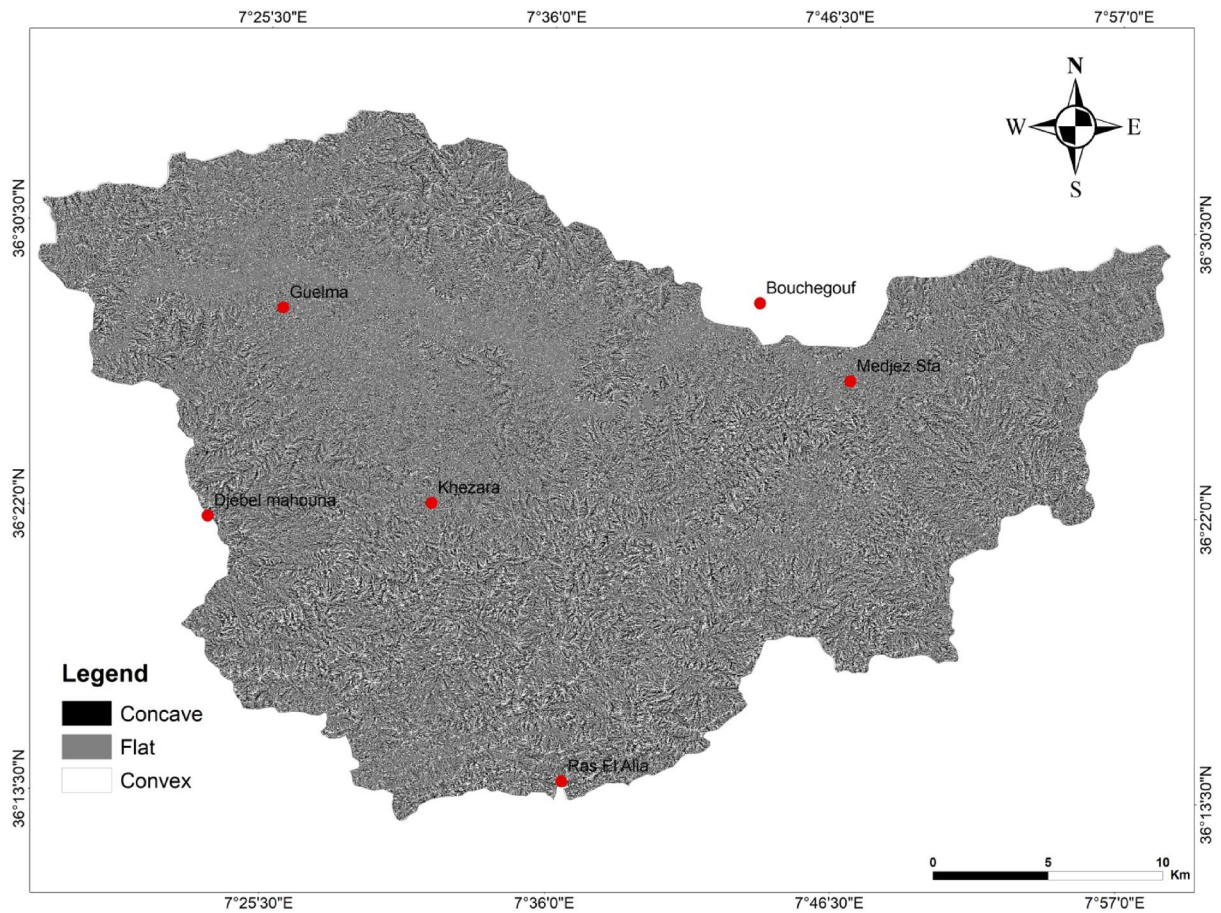


Fig. 5. Curvature plane map of the study area.

A multi-stage process model, which includes input, data processing, AHP (Analytic Hierarchy Process) multi-criteria analysis, and output, is illustrated in (Fig. 14). of the of the adopted of the adopted approach.

Results

The spatial pattern of landslide occurrences within the study area predominantly corresponds to steeply inclined slopes underlain by lithological units known for their proneness to slope failure. These include Triassic clayey-gypso-sandstone facies, unconsolidated Quaternary deposits, highly plastic Numidian clays, black marls, marly limestones, red clays, puddingstone-type sandstones, lacustrine limestones, and schistous marls. The mechanical strength and hydrogeological behavior of these formations significantly contribute to slope instability.

To quantify the relative influence of eleven key causal factors, the analytical hierarchy process (AHP) was applied using a pairwise comparison matrix (Table 3). The analysis identified slope angle as the most critical factor, assigned a weight of 0.23, followed closely by lithology (0.20) and precipitation (0.18). These results highlight the dominant role of topographic steepness and geological characteristics in controlling landslide susceptibility across the region.

The Landslide Susceptibility Index (LSI) was computed through a weighted linear combination of all factors, expressed as:

$$LSI = (0.23 \times \text{slope}) + (0.20 \times \text{lithology}) + (0.18 \times \text{precipitation}) + (0.13 \times \text{distance to faults}) + (0.08 \times \text{distance to roads}) + (0.05 \times \text{distance to wadi}) + (0.04 \times \text{aspect}) + (0.03 \times \text{altitude}) + (0.02 \times \text{plane curvature}) + (0.02 \times \text{SPI}) + (0.02 \times \text{TWI}).$$

The resulting Landslide Susceptibility Map (LSM) was classified into five categories (very low, low, moderate, high, and very high susceptibility) using the Natural Breaks method. The southern sector of the study area exhibited very high susceptibility, whereas the central and northwestern zones were predominantly characterized by low susceptibility (Fig. 15). Table 4 details the spatial distribution of susceptibility classes, revealing that approximately 41.64% of the area falls under very low susceptibility, 26.54% moderate, 21.39% high, and 10.44% very high susceptibility zones.

Model validation and interpretation

Validating the landslide susceptibility model is essential to assess its predictive performance and reliability. In this study, 70% of the inventoried landslides (352 events) were randomly selected for model training, while the remaining 30% (149 events) were used for validation. Model accuracy was evaluated using the Area Under

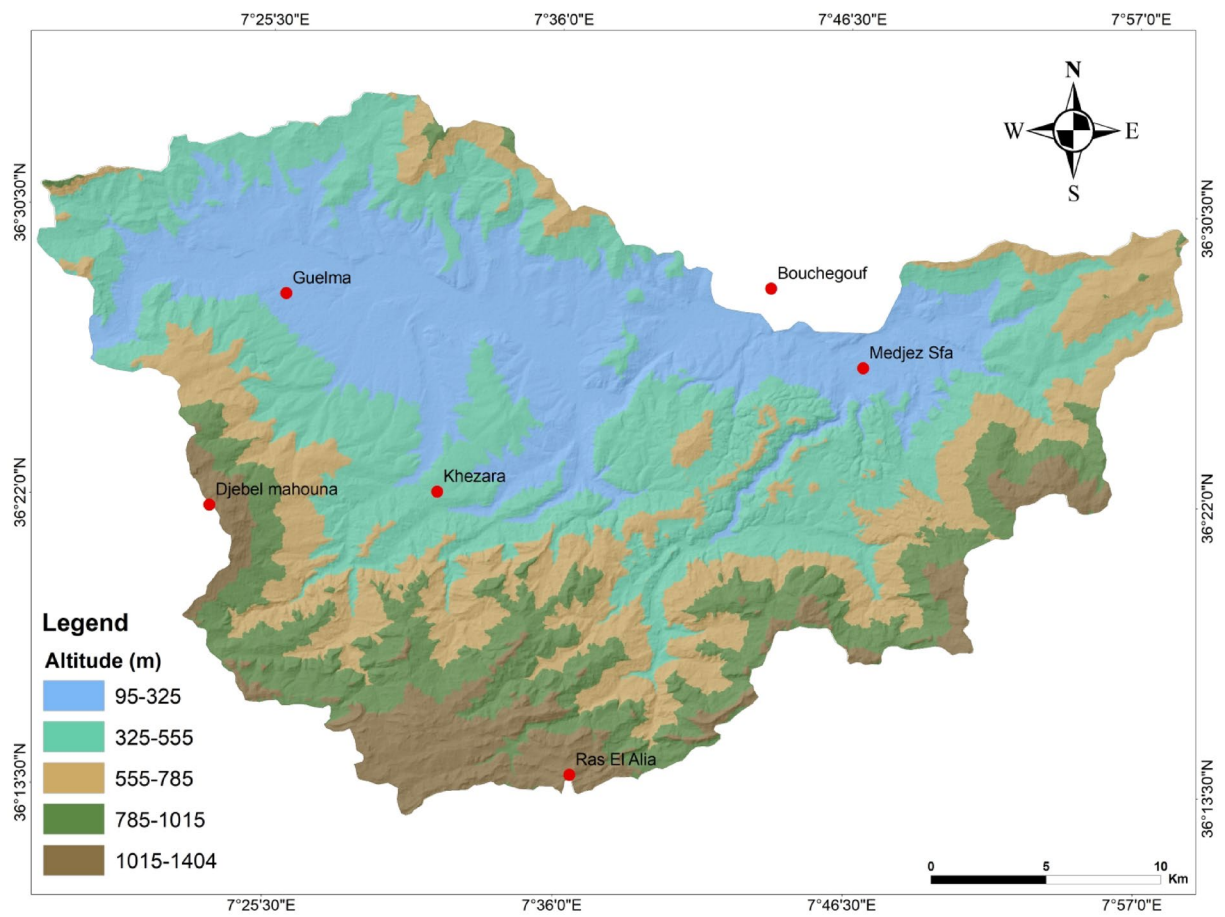


Fig. 6. Elevation map of the study area.

the Curve (AUC) derived from Receiver Operating Characteristic (ROC) analysis. The AHP model yielded an AUC of 75.10% (Fig. 16), indicating a satisfactory level of predictive capability and a good agreement between predicted susceptibility zones and observed landslide occurrences.

This result affirms the model's practical utility for supporting spatial planning and early hazard mitigation efforts, especially in data-scarce regions where expert-driven approaches are often preferred. The AHP method's structured framework, interpretability, and ability to integrate diverse geospatial factors make it well-suited for such contexts. Nonetheless, future research will explore comparative evaluations using machine learning classifiers (such as logistic regression, random forest, and support vector machines) to benchmark performance, refine susceptibility mapping, and enhance the robustness of predictions across varied geomorphological environments.

Discussion

This study's results reaffirm the paramount importance of slope angle, lithology, and precipitation in shaping landslide susceptibility, aligning well with findings from recent research in similar geomorphological contexts. The prominence of slope angle reflects the fundamental influence of gravitational forces and reduced soil cohesion on steep slopes, which inherently predispose these areas to failure. Lithological variability, especially the presence of weak clayey and marly formations, further exacerbates instability by diminishing shear strength and enhancing water retention within the soil matrix.

The significant weight assigned to precipitation underscores its role as a key landslide trigger, consistent with numerous studies linking rainfall intensity and duration to slope failures via mechanisms such as increased pore water pressure and decreased effective stress. The inclusion of distance to faults and rivers in the susceptibility model highlights the tectonic and hydrological factors that modulate landslide occurrence, corroborated by regional geomorphological analyses.

Compared to contemporary machine learning-based susceptibility models, the AHP approach offers distinct advantages in terms of interpretability and ease of application, particularly in data-scarce environments. While advanced classifiers like Random Forests and Support Vector Machines often yield higher accuracy, the transparent weighting scheme of AHP facilitates expert knowledge integration and provides a reliable baseline for susceptibility mapping.

The model's validation, with an AUC exceeding 75%, demonstrates its effectiveness in distinguishing stable from unstable areas, supporting its utility for hazard mitigation and land-use planning. This level of performance

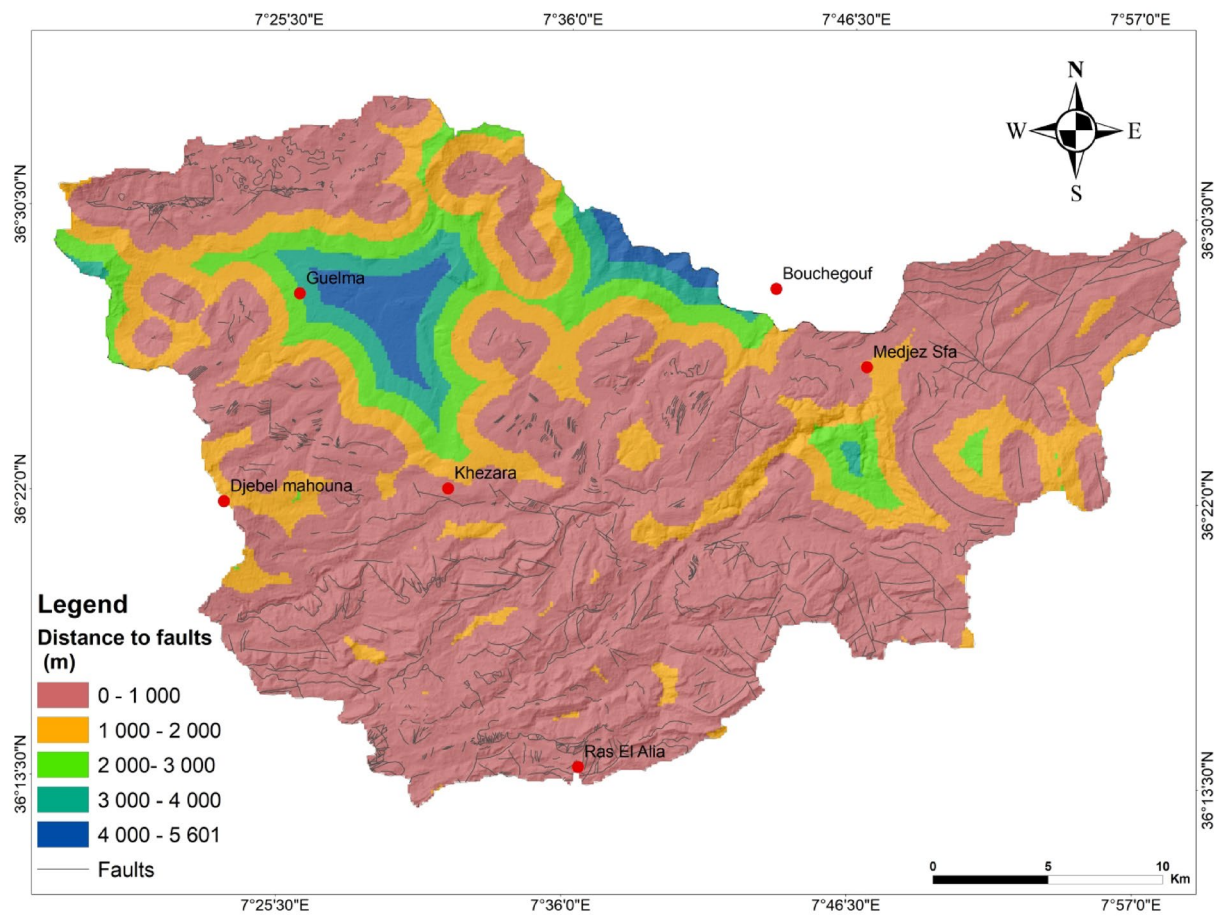


Fig. 7. Fault distance map of the study area.

is comparable to or exceeds that reported in other AHP-based studies conducted in analogous mountainous regions.

Furthermore, the combination of expert-driven AHP methodology with rigorous validation addresses common challenges in susceptibility modeling, including data limitations and factor selection biases. The resulting model offers a scientifically robust and operationally practical tool adaptable to similar mountainous landscapes worldwide.

This research advances landslide susceptibility assessment by integrating a comprehensive suite of eleven conditioning factors spanning geomorphological, geological, hydrological, climatic, and anthropogenic domains, all tailored to the local environmental context. The systematic weighting and classification framework enhances spatial resolution and improves discrimination of susceptibility zones beyond prior regional efforts.

Future work will focus on benchmarking this approach against machine learning algorithms to refine susceptibility predictions further and incorporate temporal dynamics of landslide hazards for improved risk management.

Conclusions, recommendations and outlook perspectives

This research applied a GIS-based-AHP model to assess landslide susceptibility within the Middle Seybouse watershed by integrating eleven influential factors: slope degree, slope aspect, elevation, curvature, lithology, distances to faults, rivers, and roads, alongside the TWI, SPI, and precipitation. These variables were thoughtfully selected based on their scientific significance, availability of reliable data, and appropriateness for the study's spatial resolution. The analysis involved mapping 501 landslide events, with 70% used to calibrate the model and 30% reserved for validation. The resulting susceptibility map categorized the region into five classes (from very low to very high susceptibility) with the southern and southeastern zones identified as the most prone to landslides. The model demonstrated strong predictive performance, achieving an AUC of 75.10%.

Beyond its local relevance, this study offers valuable insights for infrastructure planning, land management, and disaster risk reduction within the watershed. By integrating a diverse set of geomorphological, geological, hydrological, climatic, and anthropogenic factors within a transparent and reproducible AHP framework, the approach enhances both the reliability and interpretability of susceptibility predictions. This makes it a practical decision-support tool for engineers, planners, and policymakers aiming to mitigate landslide hazards.

On a broader scale, the methodology and findings contribute to the expanding global discourse on landslide risk management, particularly in mountainous regions increasingly vulnerable to climate change and human-

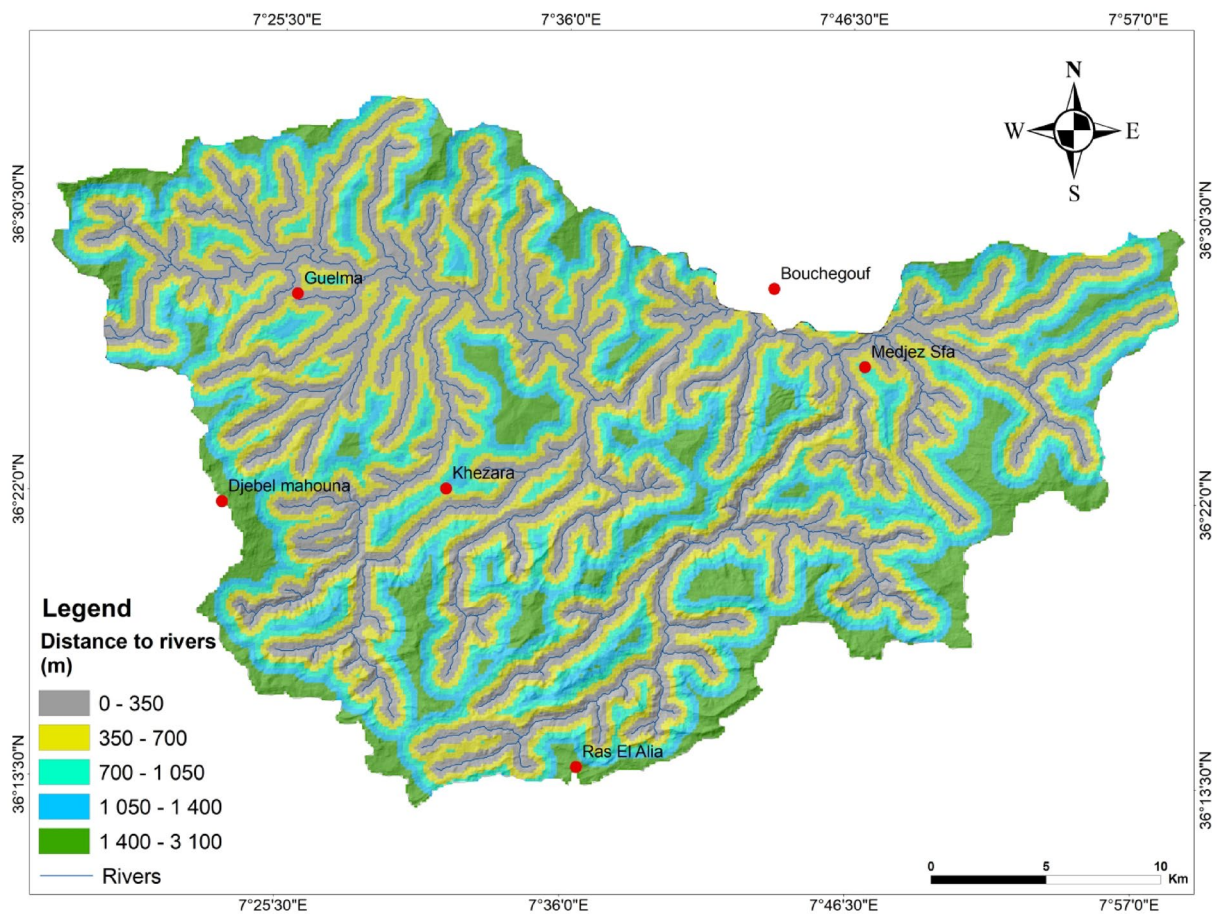


Fig. 8. Distance to rivers map of the study area.

induced pressures. The approach aligns well with international initiatives such as the Global Infrastructure Resilience Index (GIRI) and other comprehensive susceptibility assessments by emphasizing multi-factor integration and rigorous model validation. Thus, this work provides a replicable framework adaptable to similar environments worldwide, supporting global efforts toward disaster resilience and sustainable land use.

Moreover, this research underscores the value of combining expert judgment with geospatial technologies to overcome data scarcity challenges common in many regions. By refining factor weighting and classification schemes specific to local conditions, the study advances scientific understanding and improves the precision of susceptibility mapping. Its outcomes lay a foundation for future research comparing expert-driven models with machine learning techniques, fostering the evolution of landslide hazard prediction.

This study not only advances landslide susceptibility assessment at the regional level but also makes a significant contribution to the global scientific community's efforts to reduce landslide risks. Its methodological rigor and practical applicability establish a meaningful footprint in natural hazard management, highlighting the indispensable role of integrated, data-informed approaches in safeguarding vulnerable landscapes worldwide.

In our future research, we aim to broaden validation practices by incorporating a comprehensive range of performance metrics such as sensitivity, specificity, precision, recall, F1-score, and success rate curves. These metrics will provide a deeper and more balanced understanding of model performance, especially in managing the trade-offs between false positives and false negatives, which is crucial for reliable landslide hazard prediction and effective risk mitigation.

We also plan to explore and compare advanced machine learning techniques (including Random Forest, Logistic Model Trees, Support Vector Machines, and Radial Basis Function Classifiers) to harness their superior predictive power over traditional methods. Investigating hybrid models that combine these algorithms could further improve the ability to capture complex, nonlinear relationships among landslide factors and enhance overall model robustness.

Another key focus will be integrating dynamic and real-time datasets, such as rainfall, soil moisture, and land-use changes, alongside remote sensing and GIS technologies. This integration will help develop more adaptive and timely susceptibility models. By coupling machine learning with temporal analysis, we aim to enable near real-time landslide monitoring and early warning systems, thereby strengthening disaster preparedness.

Additionally, we will refine feature selection and dimensionality reduction approaches, including correlation analysis and multicollinearity testing, to streamline input variables. This will reduce noise and improve model interpretability by focusing on the most influential factors driving landslide susceptibility.

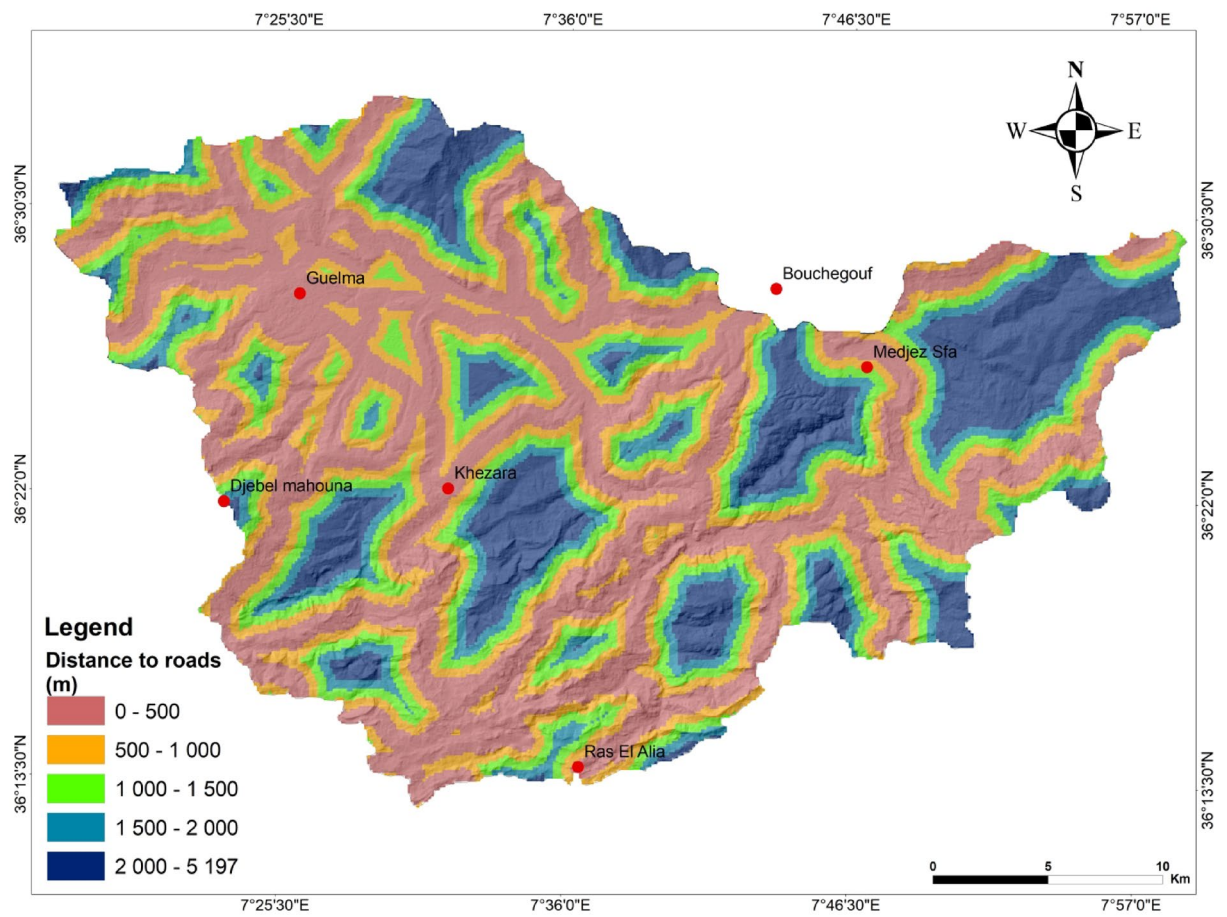


Fig. 9. Road distance map of the study area.

Finally, establishing standardized and transparent workflows with rigorous validation protocols will be a priority to ensure reproducibility and facilitate meaningful comparisons across different studies. This will support the development of globally applicable landslide susceptibility maps that contribute effectively to international disaster risk reduction and sustainable land management initiatives.

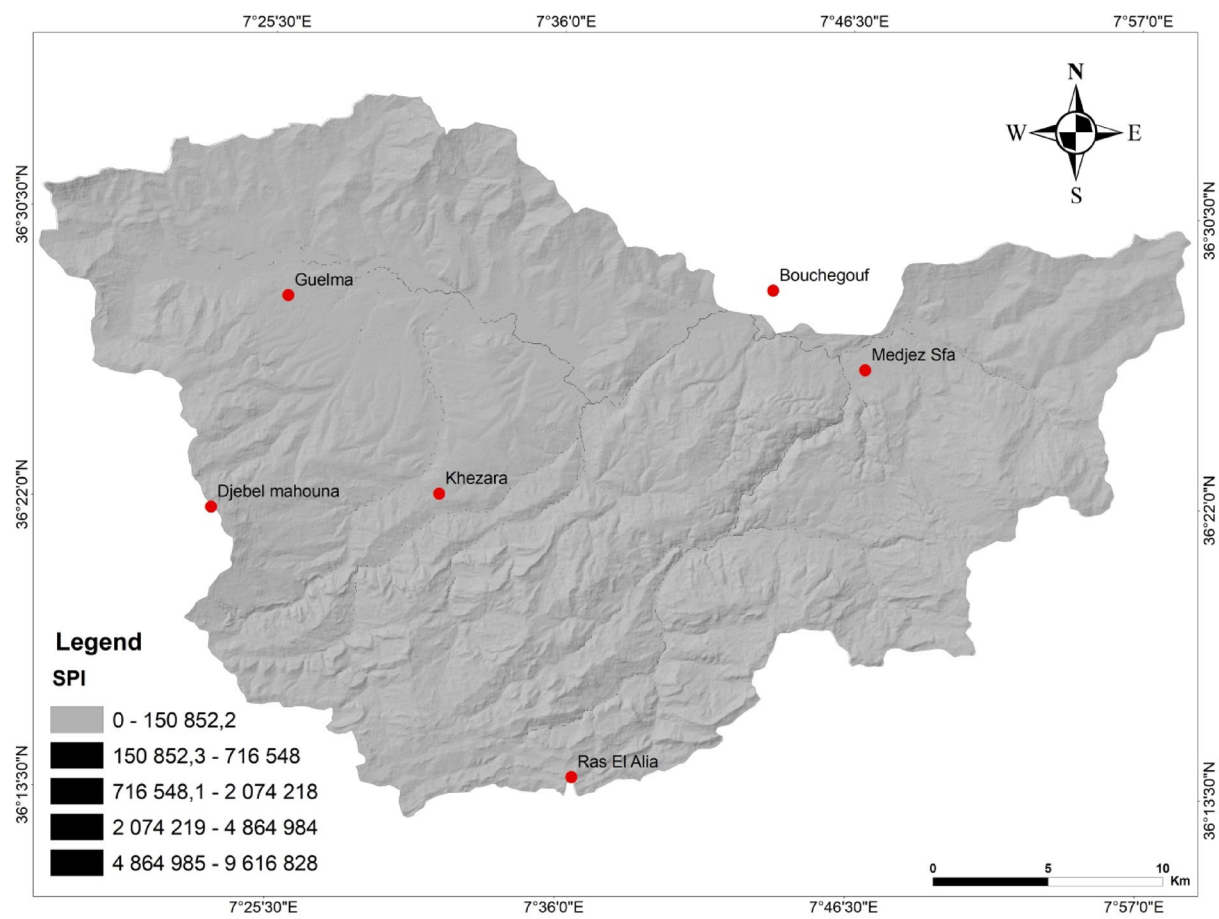


Fig. 10. Stream power index (SPI) map of the study area.

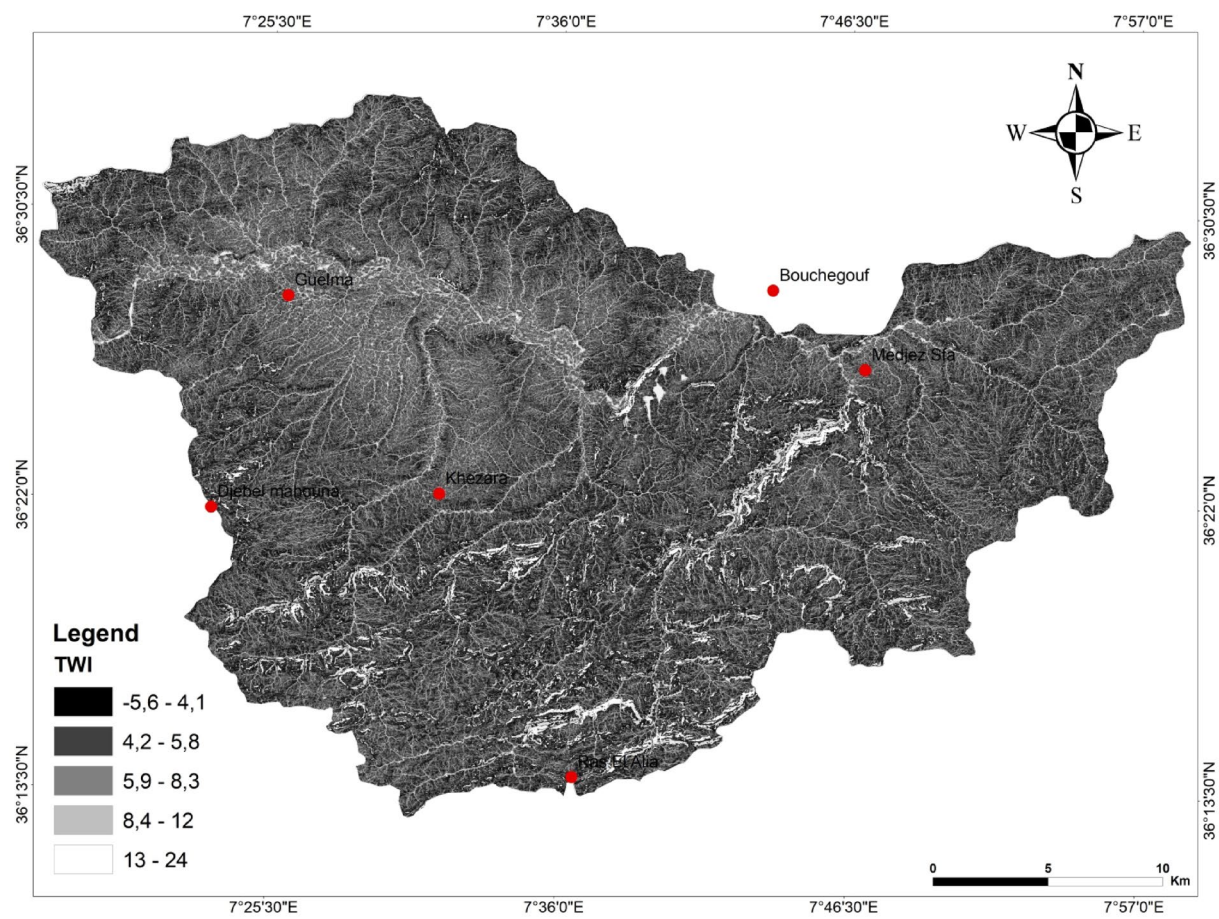


Fig. 11. Topographic wetness index (TWI) Map of the study area.

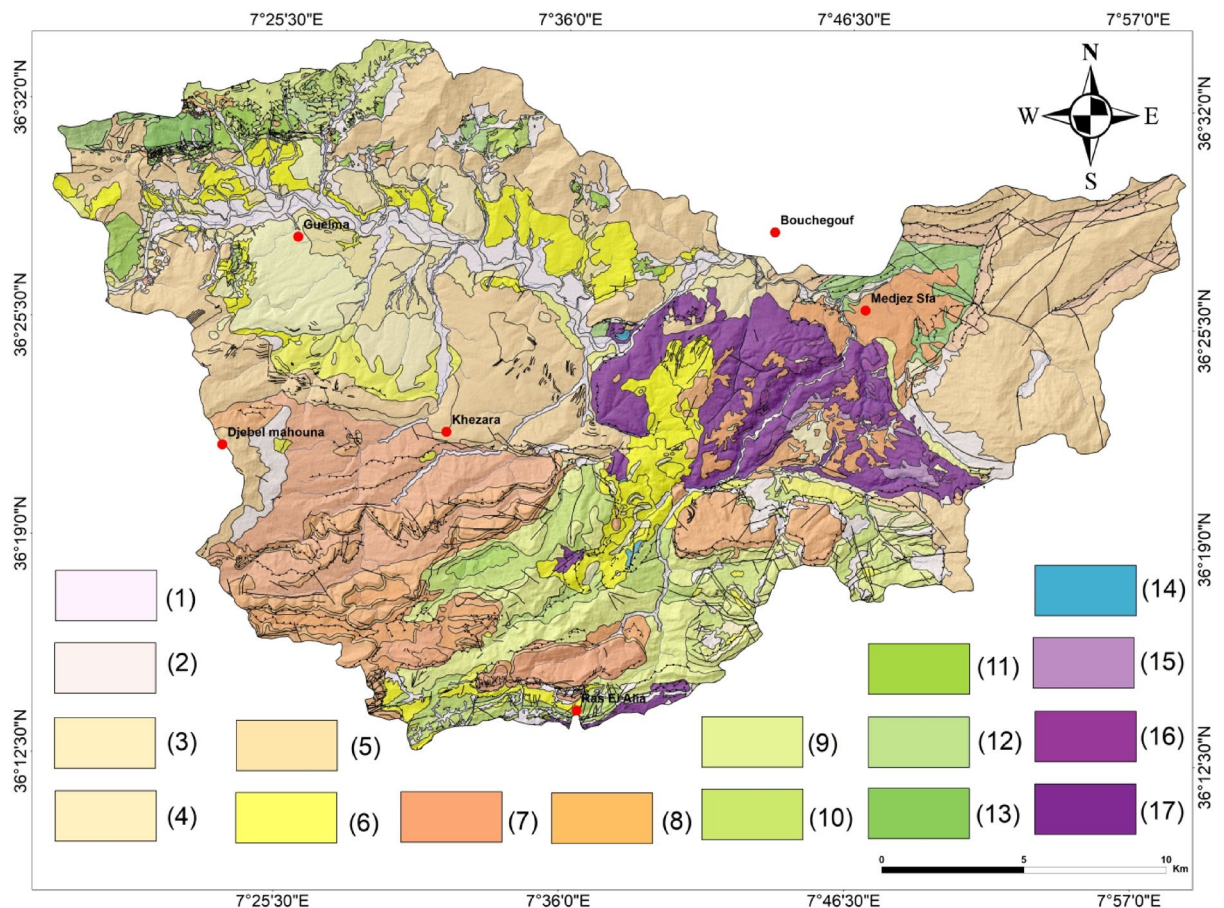


Fig. 12. lithologic map of the study area. (1) Arable land, slope formations, river alluvial deposits, edge of terraces (recent alluvial deposits), (2) sandstone scree, marly matrix scree, (3) alluvial deposits from the lower valley terraces, alluvial deposits from the middle terraces, pebbles, gravels, sands, silts, Alluvium from the high terraces of the valleys, (4) Deluvio- gravitational formations, scree slopes: breccias, gravel, sands, silts, (5) Clays, marls, siltstone intercalations of sandstone, Sandstone red clays and puddingstones, (6) Numidian clays, and sandstones (molasse of guelma-Numidian clays), (7) Marl-limestones and limestones with phosphated and bituminous flints, Limestones with nummulites, (8) Black marls, marly limestones, Red clays, sandstones of puddingstone, intercalation of marls, lacustrine limestones, (9) Limestone, sandstone limestones and marl-limestones with inoceramics, (10) Marls and gray clayey marls and past marl-limestones, (11) Schistous marls, (12) Black shale, (13) Sandstone, (14) Liassic limestone, (15) Alternation of red and bluish sandstone, clay and marl (16) Dolomites and gray dolomitic breccias, cagneules (17) Clayey-gypso-sandstone formations with fragments and blocks of various rocks.

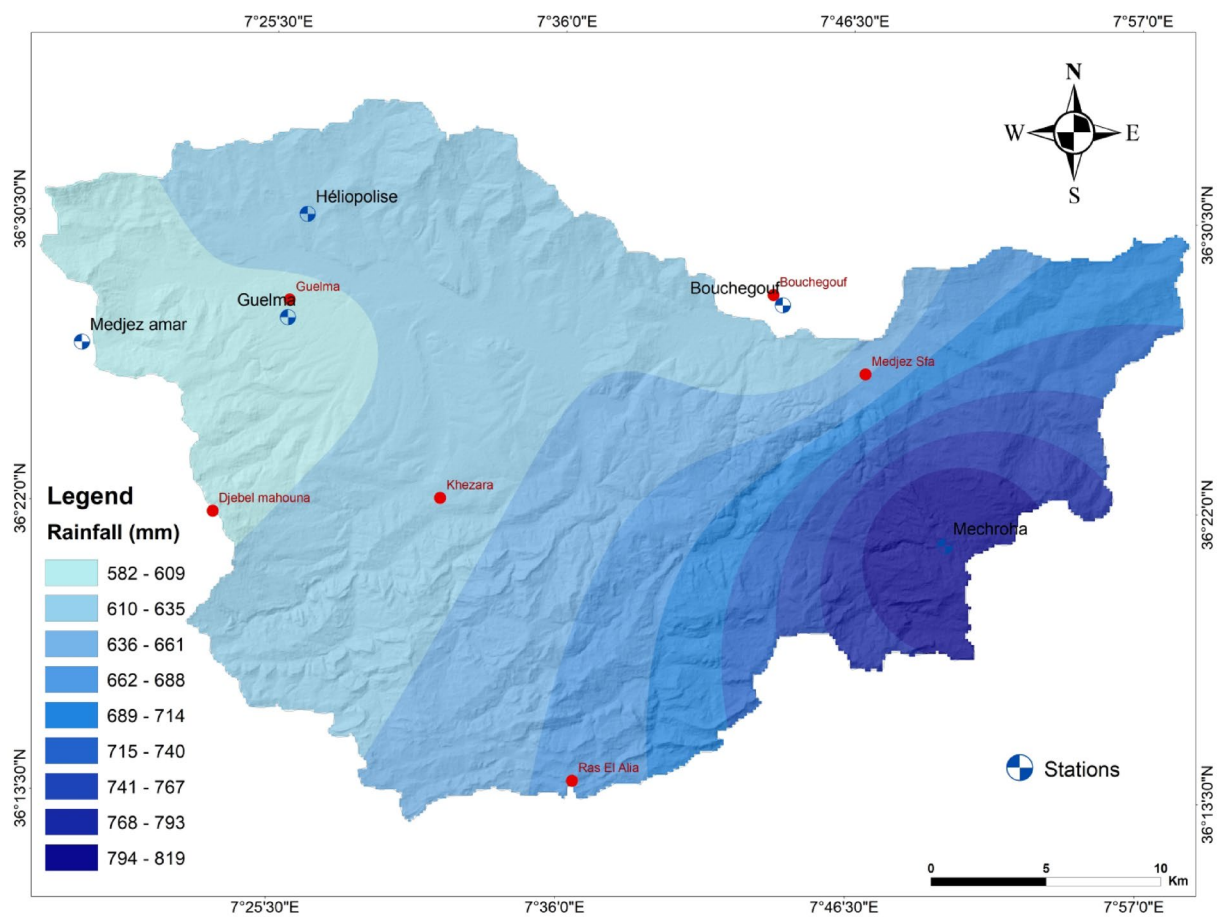


Fig. 13. rainfall map of the study area.

Conditioning factor	VIF	1/VIF
Slope angle	2.35	0.426
Lithology	3.42	0.292
Precipitation	2.87	0.348
Distance to faults	1.89	0.529
Distance to roads	2.13	0.469
Distance to rivers	1.76	0.568
Slope aspect	1.21	0.826
Altitude	4.02	0.249
Plane curvature	3.67	0.272
SPI	2.48	0.403
TWI	4.87	0.205

Table 2. Multicollinearity diagnostics for the landslide-conditioning factors.

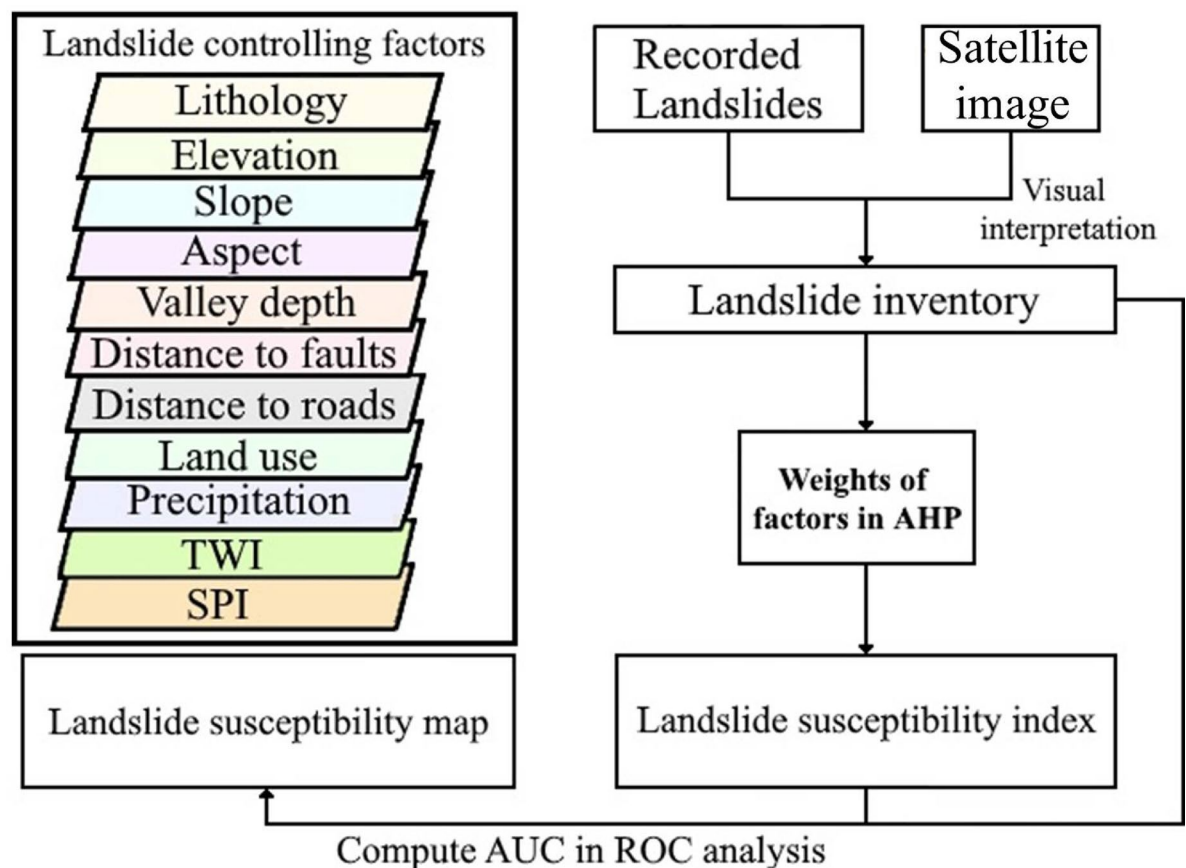


Fig. 14. Methodological flowchart of the adopted approach.

Influencing factors	Pair-wise comparison matrix											Weight
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	
(1) Slope angle	1	2	2	3	4	6	7	7	7	6	6	0.23
(2) Lithology		1	2	3	5	5	6	6	7	7	4	0.20
(3) precipitation			1	3	5	5	6	6	7	6	7	0.18
(4) Distance to faults				1	3	6	5	7	7	6	7	0.13
(5) Distance to roads					1	3	3	5	4	6	5	0.08
(6) Distance to rivers						1	2	3	4	4	4	0.05
(7) Slope aspect							1	3	4	4	5	0.04
(8) Altitude								1	3	3	3	0.03
(9) Curvature plane									1	3	4	0.02
(10) SPI										1	2	0.02
(11)TWI											1	0.02

Table 3. Pairwise comparison matrix and corresponding factor weights in AHP.

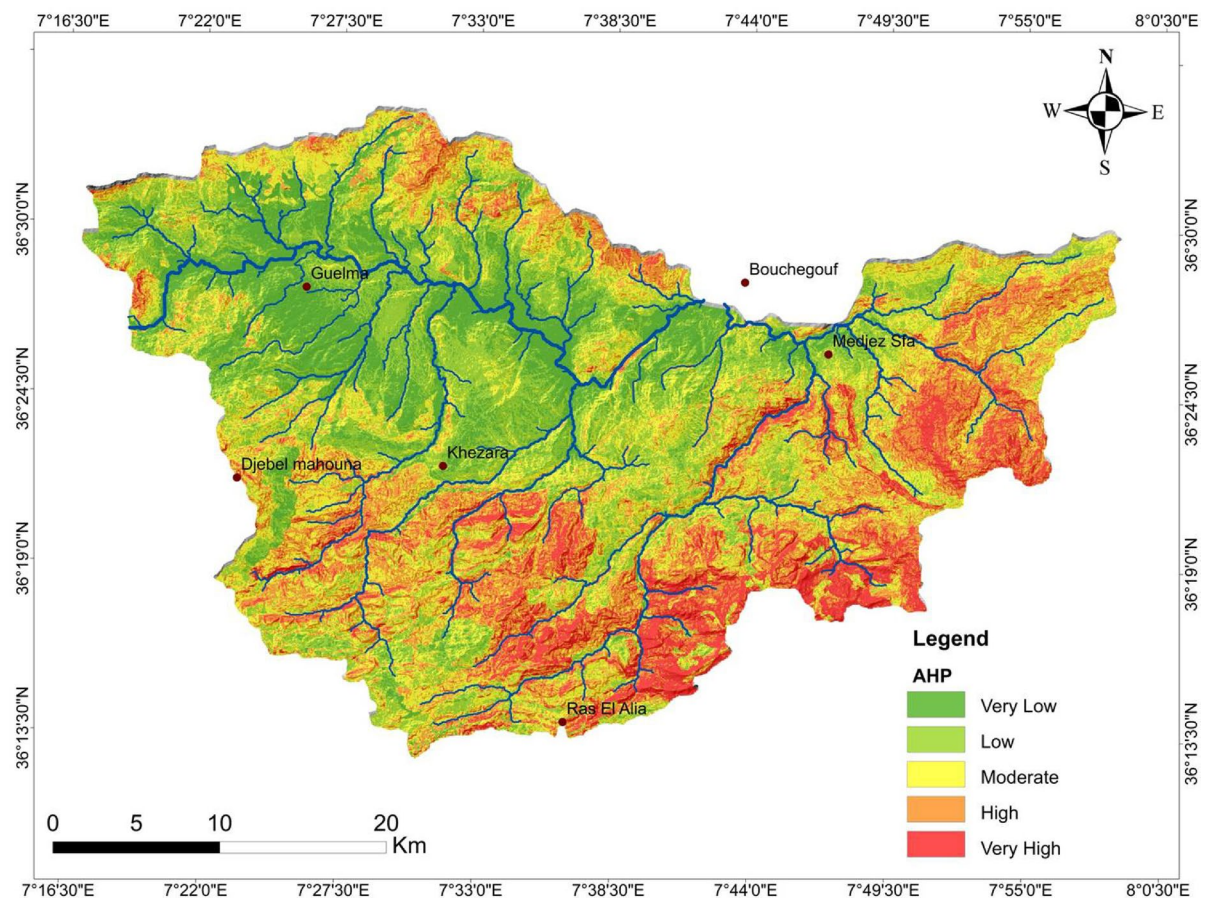


Fig. 15. Landslides susceptibility map of the study area.

Susceptibility class	AHP method	
	% area covered	% landslide area covered
Very low	16,115	1743
Low	25,523	7982
Moderate	26,537	22,844
High	21,386	37,339
Very high	10,440	30,092

Table 4. The five susceptibility zones.

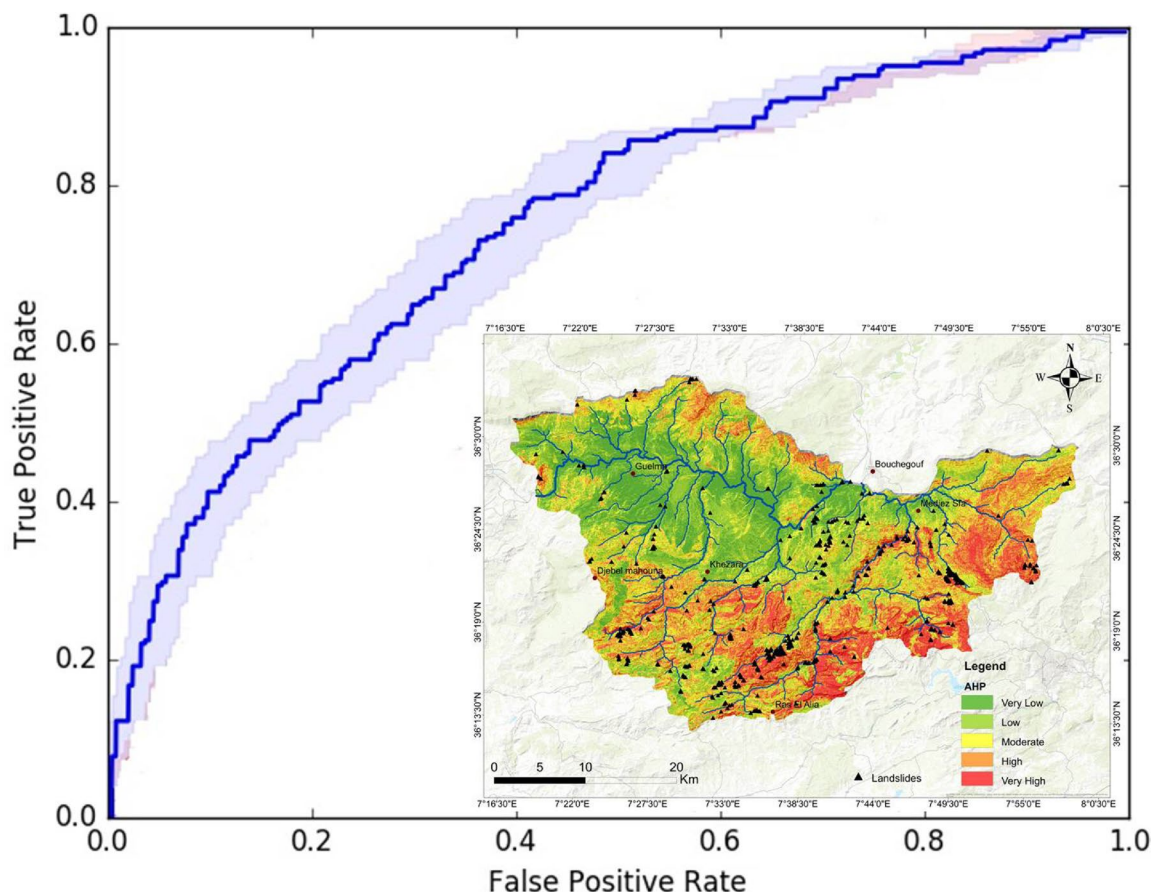


Fig. 16. The ROC curve of the retained model validation.

Data availability

All data generated or analyzed during this study are included in this published article. For additional data requests, please contact the corresponding author, who will provide access upon request.

Received: 2 April 2025; Accepted: 7 July 2025

Published online: 16 July 2025

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Acknowledgements

Princess Nourah bint Abdulrahman University Researchers Supporting Project number (PNURSP2025R911), Princess Nourah bint Abdulrahman University, Riyadh, Saudi Arabia.

Author contributions

Conceptualization, K.Z. and K.F.; methodology, K.F.; software, F.Z.; validation, K.F. and K.Z.; formal analysis, A.A.; investigation, F.Z.; resources, S.A.; data curation, F.A.; writing original draft preparation, K.Z.; writing review and editing, K.F.; visualization, E.A.; supervision, J.A.; project administration, K.Z.; funding acquisition, K.F. All authors have read and agreed to the published version of the manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to K.Z.

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